



SIOP ML Competition 2026

Team goforit





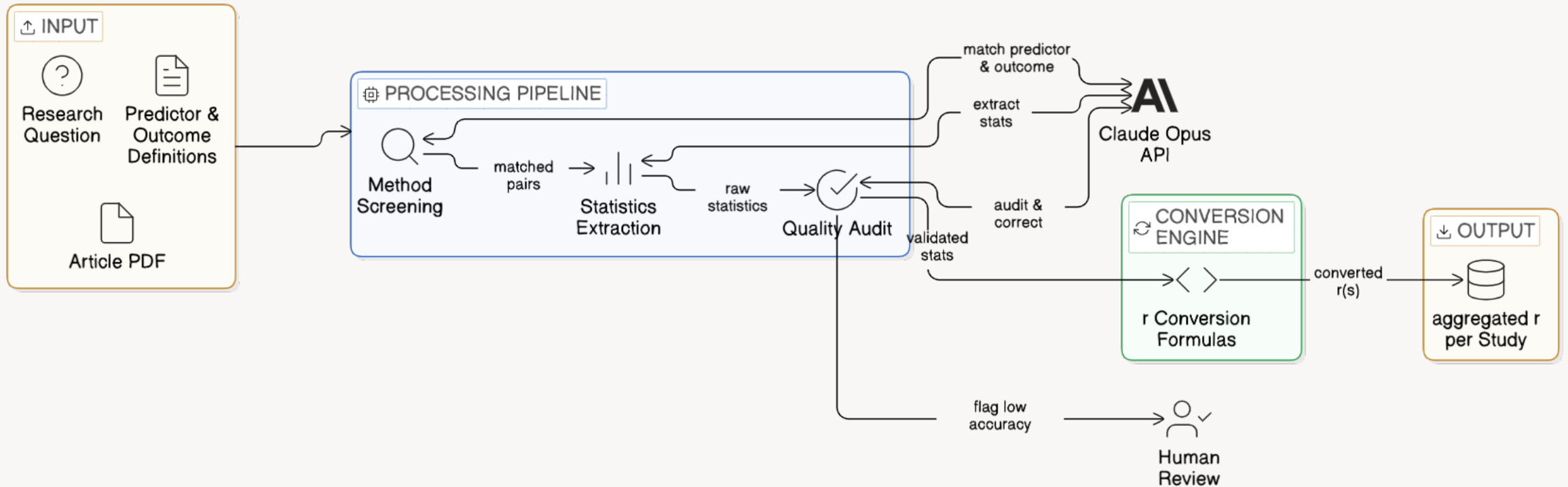
University of Minnesota

Team Members

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Pipeline Overview

A 3-step LLM-driven extraction pipeline



I-O Expertise as Pipeline Architecture

THE PROBLEM

Construct names might lie

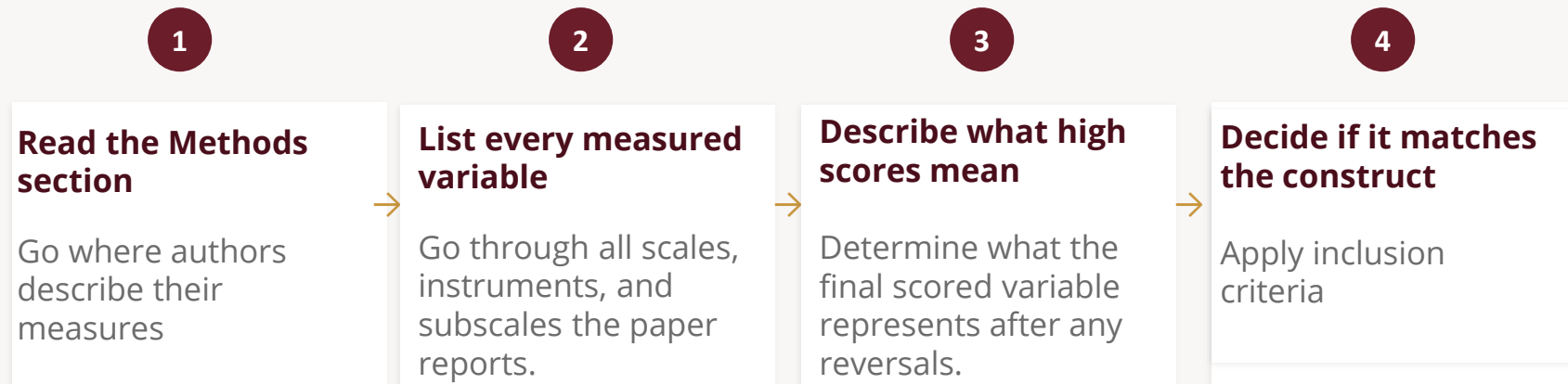
"Burnout"

can appear as: cynicism, reduced accomplishment, emotional burnout

A search for "burnout" might miss these

THE SOLUTION

Ask the LLM to act like a trained coder



I-O Expertise as Pipeline Architecture

- LLMs might be unreliable at math
 - Solution: use deterministic conversion formulas
 - t-tests, F-ratio, Cohen's d, odds ratio, chi-square, contingency tables
 - Extract based on level priority: bivariate r > derivable bivariate stats
 - Stop at the first level that yield valid statistic



Cost vs. Pipeline Simplicity

- More Steps $\neq$ Better Results
 - Earlier versions:
 - v1: pdfplumber/PyMuPDF to parse PDFs into text locally -> send text to Claude API
 - v2: rendered each pdf as image -> sent each page image to a VLM for markdown conversation
 - Final version: Send pdf directly to Claude API



Sign Correction is the Hidden Hard Problem

Negative-pole is not just about the construct name (e.g., WLB could be measured by a conflict/harmony scale)

LAYER 1 — SCREENING

Paper measures:
"Work-Family Conflict Scale"

→
LLM reads
Methods

High = more conflict
= worse work-life balance
→ **negative_pole = true**

vs.

Same construct, different scale:
"Work-Life Harmony Scale"
→ **negative_pole = false**

negative_pole depends on the instrument, not the construct name. Only reading the Methods section reveals which direction.

LAYER 2 — CONVERSION

Apply sign corrections in order

Social Support → Work-Life Balance (conflict scale) $r = -0.25$, direction = "negative", reverse_coded = true

✓ **Correct path**



✗ **Double-flip trap (if direction check NOT skipped)**



Limitations & Future Directions

LIMITATIONS

Non-Deterministic Outputs Across Runs

Same pdf can produce slightly different extractions on different runs

No Access to Supplemental Materials

When key statistics are in online supplements or separate appendices, the pipeline can only flag their absence it cannot retrieve or process them.

Cost

\$1.5-\$2 per pdf

FUTURE DIRECTIONS

Repeated Sampling with Aggregation

Run the extraction pipeline multiple times per paper (3-5 passes) and compare outputs.

External Materials Retrieval

Integrate a web retrieval step that activates when the pipeline detects a reference to supplemental materials

Explore Other Models

Combination of smaller models and local processes





Domain knowledge is still important!

Thank you!

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